Text analytics for Business Insights: Develop a models using NLP frameworks such as spaCy, BERT, or RoBERTa for analyzing customer reviews/comments. Implement advanced text analytics techniques like Aspect-based Sentiment analysis, and Topic modelling using LDA or BERTopic

```
import pandas as pd
import spacy
import re
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
df = pd.read_csv('/content/drive/MyDrive/SMA_DATASET/merged_comments.csv')
```

```
nlp = spacy.load('en_core_web_sm')

def clean_text(text):
    text = str(text).lower()
    text = re.sub(r'\s+', ' ', text)
    text = re.sub(r'http\S+', '', text)
    text = re.sub(r'[^a-zA-Z0-9\s]', '', text)
    doc = nlp(text)
    tokens = [token.lemma_ for token in doc if not token.is_stop]
    return ' '.join(tokens)

df['Cleaned_Comment'] = df['Comment'].apply(clean_text)
print(df[['Comment', 'Cleaned_Comment']].head())
```

```
                                             Comment  \
0  I'm creating an infinite canvas that has all y...
1  Man...If you built this for large mainframe co...
2  On a smaller scale it reminds me of the origin...
3  Cool. My long-term vision for software develop...
4  Sometimes I daydream that we moved beyond text...

                                     Cleaned_Comment
0  m create infinite canvas organization code doc...
1  manif build large mainframe codebase think out...
2  small scale remind original concept light tabl...
3  cool longterm vision software development new ...
4  daydream move text file format uuid lineversio...
```

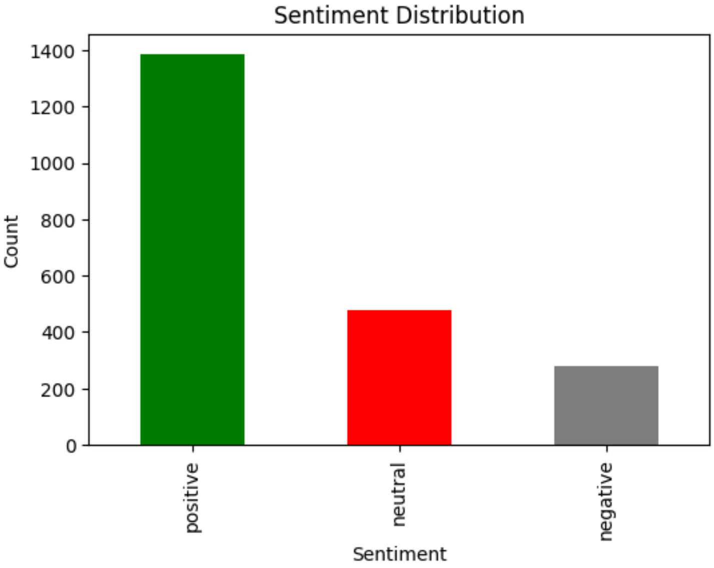
```
!pip install vaderSentiment
```

Show hidden output

```
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

# Initialize sentiment analyzer
sia = SentimentIntensityAnalyzer()

# Function to get sentiment score
def get_sentiment(text):
    sentiment = sia.polarity_scores(text)
    return 'positive' if sentiment['compound'] > 0.05 else 'negative' if sentiment['compound'] < -0.05 else 'neutral'

# Apply sentiment analysis
df['Sentiment'] = df['Cleaned_Comment'].apply(get_sentiment)

# Display sentiment counts
print(df['Sentiment'].value_counts())

# Plot sentiment distribution
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
df['Sentiment'].value_counts().plot(kind='bar', color=['green', 'red', 'gray'])
plt.xlabel('Sentiment')
plt.ylabel('Count')
plt.title('Sentiment Distribution')
```

```
plt.show()
```

```
Sentiment
positive    1388
neutral      476
negative     280
Name: count, dtype: int64
```




```
!pip install bertopic
```

Show hidden output

```
from bertopic import BERTopic
import matplotlib.pyplot as plt

# Train BERTopic model
topic_model = BERTopic()
topics, probs = topic_model.fit_transform(df['Cleaned_Comment'])

# Add topics to DataFrame
df['Topic'] = topics

# Show top 10 topics
print(topic_model.get_topic_info().head(10))

# --- Visualization ---

# 1️⃣ *Bar Chart of Top Topics*
topic_model.visualize_barchart(top_n_topics=10)

# 2️⃣ *Topic Distribution*
topic_model.visualize_distribution(probs)

# 3️⃣ *Word Cloud for Topics*
topic_model.visualize_term_rank()

# 4️⃣ *Topic Similarity Heatmap*
topic_model.visualize_heatmap()
```

```
   Topic  Count                           Name  \
0     -1    349       -1_not_dreamweaver_app_work
1      0     90          0_speed_drive_road_car
2      1     66  1_language_parser_compiler_write
3      2     58    2_video_language_learn_learner
4      3     45          3_thank_hey_sorry_read
5      4     42      4_serial_usb_port_hardware
6      5     38       5_log_monitoring_set_vacuum
7      6     37        6_code_canvas_figure_novel
8      7     36           7_api_apis_agent_aiml
9      8     36   8_ingredient_grocery_food_track

                                      Representation  \
0  [not, dreamweaver, app, work, ve, use, bit, fr...
1  [speed, drive, road, car, lane, people, limit,...
2  [language, parser, compiler, write, zig, imple...
3  [video, language, learn, learner, vocabulary, ...
4  [thank, hey, sorry, read, thosereply, uareply,...
5  [serial, usb, port, hardware, wasm, upse, insp...
6  [log, monitoring, set, vacuum, ecu, email, too...
7  [code, canvas, figure, novel, view, etc, infin...
8  [api, apis, agent, aiml, semantic, graph, rela...
9  [ingredient, grocery, food, track, eat, recipe...

                                 Representative_Docs
0  [m fashion nerd ve work outfit track app 4 yea...
1  [ironically low ground fast feel   drive sport...
2  [parser combinator library m write tool static...
3  [personal project gitmentor   300 user far    n...
4  [hey thank let knowreply, hey sorry guess user...
5  [m work catalog open source hardware designswh...
6  [meapparently find set monitoring problematic ...
7  [m create infinite canvas organization code do...
8  [thank appreciate yes software api utility com...
9  [celiac m make database single label glutenfre...
```